# Cell Detection: Edge Detection & Hough Transform

Computer Vision and Imaging [06 32578] — Mazen Salama 262805

## 1. Aim

The aim of this assignment for tasks 1-2 is to investigate automated edge detection methods (Sobel, Prewitt, Laplacian) applied to 3 HEp-2 cell images which are used in the diagnosis of autoimmune diseases. The goal is to help laboratories as manual interpretation of these images varies between experts and is subjective. Task 2 then applies the more advanced Canny detector and investigates how different threshold values affect the results. Task 3 applies the Hough Transform as covered in the Week 4 lectures to detect straight lines in a real world architectural image of University of Birmingham Dubai Campus and explores how parameter tuning affects detection accuracy.

## 2. Methods

I first converted all three cell images to grayscale using skimage's rgb2gray function. This is important as edge detectors work on intensity changes not colour, so colour information is not needed. After that I applied a Gaussian blur with a 5x5 kernel and sigma of 1 using OpenCV's GaussianBlur before running any edge detection. This removes noise from the images so the detectors dont pick up random pixels as edges. This is especially important for the Laplacian as it is a second order derivative and much more sensitive to noise than Sobel and Prewitt — without blurring it would detect noise everywhere instead of actual cell boundaries.

For Task 1 I implemented all three detectors using 3x3 kernel definitions and scipy.signal.convolve2d, the same approach used in the lab sessions. For Sobel and Prewitt I applied two kernels on each image, one for horizontal changes and one for vertical, then combined them using $\sqrt{(Gx^2+Gy^2)}$ to get the edge strength at each pixel. The difference is that Sobel has a weight of 2 in the centre which makes it slightly less sensitive to noise than Prewitt which has equal weights. For the Laplacian I used a single kernel with -8 in the centre and 1s around it which detects edges in all directions at once. For Task 2 I chose Canny which works in 4 stages — Gaussian smoothing, gradient computation, non-maximum suppression to thin edges to one pixel wide, and hysteresis thresholding with t2=2xt1 as required. I tested three threshold pairs to investigate the effect. For Task 3 I applied the Hough Transform using the Canny edge map as input. The Hough Transform works by having each edge point vote for possible line parameters and lines are detected where the most votes accumulate. I used cv2.HoughLinesP and tuned the parameters step by step to reduce false detections.

## 3. Results & Discussion

Figure 1 shows Sobel applied to all three images. The results look clean and cell boundaries are clearly visible with minimal noise. Comparing to the ground truth, Sobel detects most cell outlines but the edges are slightly thicker than the ground truth which shows thin precise single pixel boundaries. Sobel also picks up internal cell structures which do not appear in the ground truth — this is because it detects any brightness change it finds while the human expert only marked the outer boundary. I also noticed in the top right corner of Image 1 that Sobel detected an edge visible in the original image but not included in the ground truth, showing that the human annotator made a subjective judgement/or maybe human error while Sobel detects everything mathematically.

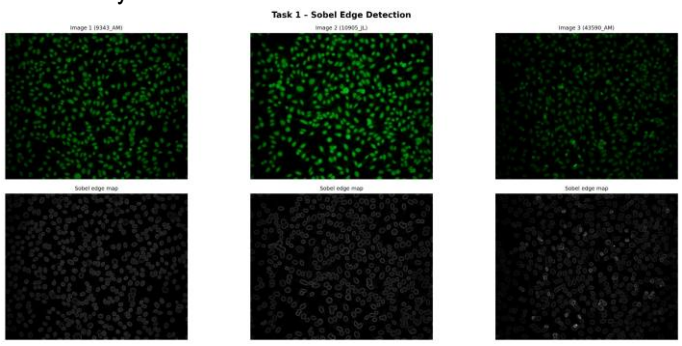



*Figure 1 – Sobel: all three images. Top row: originals. Bottom row: Sobel edge maps.*

Figure 2 shows Prewitt results which look almost identical to Sobel — it is very hard to tell the difference just by looking at them. Both detectors work the same way and the only difference is Sobel has a weight of 2 in the centre while Prewitt uses equal weights, but in practice on these images that difference is not visible. Prewitt has the same limitations as Sobel with slightly thicker edges and internal cell structures also being picked up.

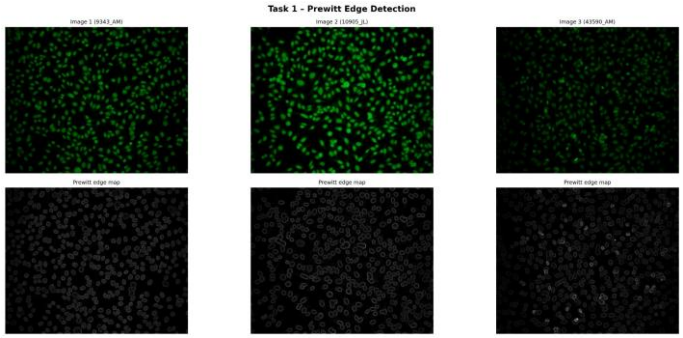



*Figure 2 – Prewitt: all three images. Results visually identical to Sobel.*

Figure 3 shows the Laplacian which looks very different. The background has a grey noisy texture all over it and inside the cells themselves there is also a lot of grey texture which does not appear in the ground truth at all. Image 3 is the worst where cells almost merge into each other. Even with Gaussian blur applied before it still picks up way more than just the cell boundaries making it the least suitable detector out of the three for these images.

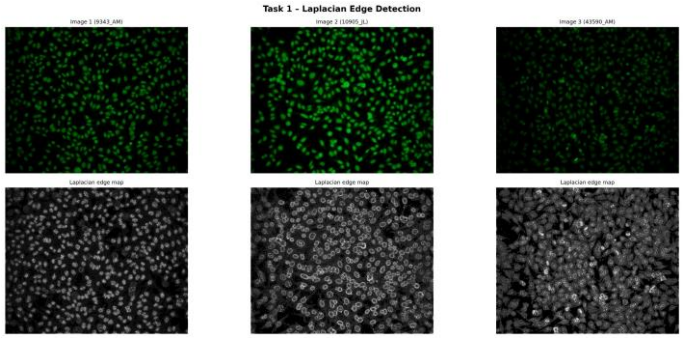



*Figure 3 – Laplacian: all three images. Significantly more noise than Sobel and Prewitt.*

Figure 4 shows Canny across three threshold pairs. Canny produces very thin clean edges with a completely black background which looks the most similar to the ground truth in terms of edge quality, and it also detected the internal edges inside the cells very well. I tested three threshold pairs — at (t1=0.05, t2=0.1) the most cells were detected and the result was closest to the ground truth, at (t1=0.1, t2=0.2) edges got cleaner but more cells disappeared, and at (t1=0.2, t2=0.4) almost nothing was detected. However overall I think Sobel and Prewitt performed better for these images because Canny misses a lot of cells on the darker side where fluorescence is weaker — those cells are clearly in the ground truth but Canny ignores them due to strict thresholding. In medical imaging missing cells is a bigger problem than having slightly thicker edges so Sobel and Prewitt are the more reliable choice here.

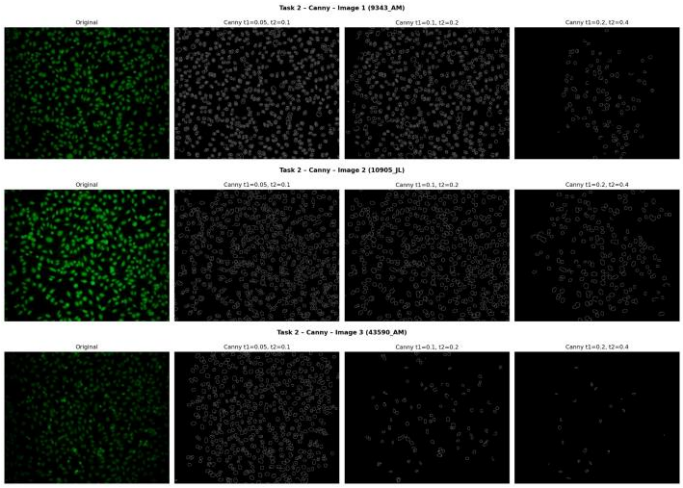



*Figure 4 – Canny threshold comparison across Images 1, 2, 3. Columns: t1=0.05/t2=0.1, t1=0.1/t2=0.2, t1=0.2/t2=0.4.*

Figure 5 shows the full Hough Transform experiment. The top strip shows that I first used Canny to extract the edges using sigma=2.0 which applies a stronger blur to reduce noise from the trees and plants. The remaining four strips show the results as I tuned the parameters step by step. The first attempt with threshold=80 and minLineLength=60 detected 1075 lines which was way too many. I raised the threshold to 150 and minLineLength to 100 bringing it down to 388, then further to threshold=200 and minLineLength=150 getting it to 195. The final parameters with threshold=250, minLineLength=200 and maxLineGap=20 gave 13 lines correctly detecting the main horizontal floor levels of the building and parking bay lines. Some real lines like window frames were lost during this process which shows the main challenge — the stricter the parameters the fewer false lines but also more real ones get missed.

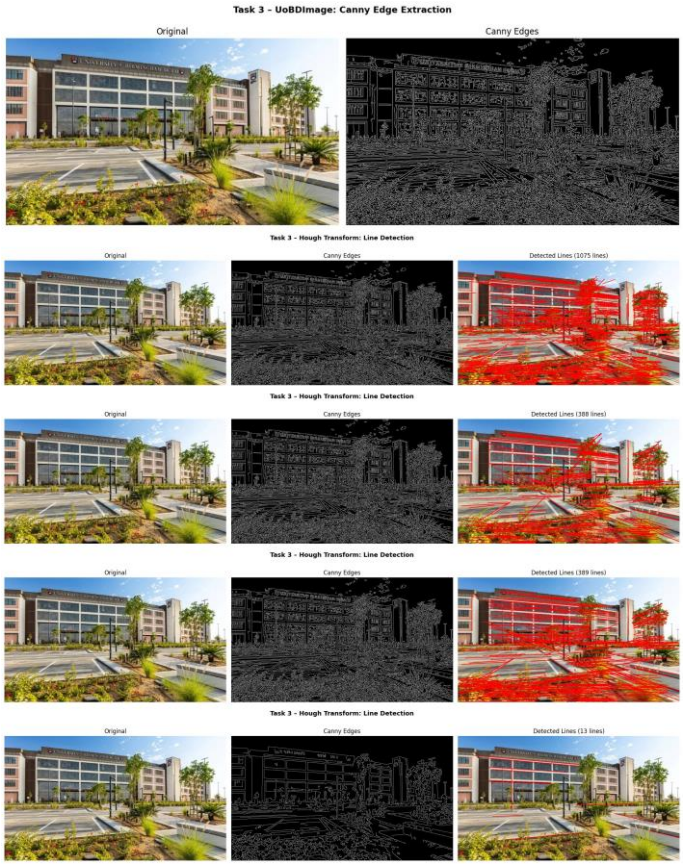



*Figure 5 – Task 3: Canny extraction (top) then Hough results with 1075, 388, 195 and 13 detected lines.*

## 4. Conclusion

To summarise, Sobel and Prewitt both performed well and produced the clearest results out of the three Task 1 detectors. They are very similar and it is hard to tell the difference — Sobel is slightly better in theory because of its centre weighted kernel but in practice on these images they look the same. The Laplacian was the worst as it picked up too much noise and internal cell texture making it not suitable without extra processing. For Task 2 Canny produced the thinnest and cleanest edges and detected internal cell structures very well, but I think Sobel and Prewitt performed better overall because Canny missed a lot of darker cells clearly visible in the ground truth. In medical imaging missing cells is a bigger problem than slightly thicker edges so Sobel and Prewitt are the more reliable choice. For the Hough Transform I managed to detect the main structural lines after tuning parameters step by step but making parameters stricter reduces false detections while also causing real lines to be missed. Overall I would recommend Sobel or Prewitt for detecting all cell boundaries in these images as they gave the most complete results.